The Journal of Open Source Software

VibeBench: An Automated Framework for the Holistic Evaluation of LLM-Generated Code

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DOI: 10.xxxxxx/draft

Software

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Submitted: 01 January 1970
Published: unpublished

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Summary

The rapid integration of Large Language Models (LLMs) into the software development lifecycle has created a critical need for evaluation frameworks that extend beyond basic functional correctness. Traditional benchmarks, such as HumanEval or MBPP, primarily measure a model's ability to pass unit tests (functional correctness). However, they often overlook essential software engineering attributes like code maintainability, structural complexity, and operational resource efficiency.

VibeBench is an automated, extensible Python framework designed to fill this gap. It provides a holistic evaluation of LLM-generated code by integrating static quality heuristics with sandboxed dynamic execution. By walking the Abstract Syntax Tree (AST), VibeBench quantifies Halstead complexity metrics and identifies non-standard "bad practices"—such as ghost comments, hardcoded credentials, and documentation gaps—that are frequently found in AI-synthesized outputs but missed by traditional testers.

Simultaneously, the framework manages a secure lifecycle for generated scripts using Unix-based resource limiting (CPU time and memory address space) to measure runtime stability and latency. VibeBench is built for AI researchers, software auditors, and developers who need to quantify the "technical debt" introduced by autonomous agents. By grounding AI performance against a formalized Human Baseline, it provides the necessary metrics to determine if AI-generated code is truly production-ready.

VibeBench is designed for three primary user groups. AI researchers studying model evolution can use it to conduct longitudinal studies comparing how code quality changes across model versions. Software engineering teams evaluating AI coding assistants for production deployment can use it to audit technical debt before it accumulates. Benchmark designers can use it as a foundation for extending evaluation beyond functional correctness. The framework is released under the MIT License and designed to be extensible — new models, tasks, and heuristics can be added without modifying the core pipeline.

Statement of need

Current evaluation methodologies for Large Language Model (LLM) generated code, such as HumanEval (Chen & others, 2021) and MBPP (Austin & others, 2021), primarily focus on functional correctness through unit testing. While these benchmarks are effective at determining if a model can solve a specific problem, they fail to audit the structural integrity and production-readiness of the resulting code. In a production environment, code that "works" but is overly complex, undocumented, or resource-inefficient creates significant technical debt and security risks.

VibeBench addresses this gap by providing a toolset that treats AI-generated code as software artifacts rather than just mathematical solutions. There is a documented "documentation crisis"

in LLM outputs, where models frequently omit docstrings and internal comments required for maintainability. Furthermore, existing benchmarks rarely account for the correlation between high structural complexity and runtime instability.

VibeBench allows researchers and software auditors to:

- **Quantify Technical Debt:** Measure Halstead complexity and documentation coverage to predict long-term maintenance costs.
- **Audit Security and Best Practices:** Detect "AI-isms" like ghost comments or hardcoded credentials using AST-based heuristics.
- **Benchmark Operational Parity:** Compare AI performance against a formalized human baseline in a resource-constrained sandbox.

By providing these capabilities in an open-source, modular format, VibeBench empowers researchers to conduct large-scale longitudinal studies on model evolution, benchmark the energy efficiency of AI-generated algorithms for Green IT research, and build "AI-Audit" pipelines that ensure autonomous agents adhere to human-centric documentation and security standards. This enables the scientific community to move toward a more holistic understanding of AI performance that prioritizes long-term software sustainability over short-term functional success.

State of the field

Several benchmarks and frameworks currently exist for evaluating the code generation capabilities of Large Language Models (LLMs). The most prominent among these are HumanEval (Chen & others, 2021) and MBPP (Mostly Basic Python Problems) (Austin & others, 2021). These benchmarks establish functional correctness by measuring the pass@k metric—a statistical representation of whether a model can produce at least one solution that passes a provided suite of unit tests. Other tools, such as CodeSearchNet (Husain & others, 2019), provide large-scale datasets for code retrieval and summarization but are not designed for direct execution-based benchmarking.

Despite their widespread adoption, these tools share a common limitation: they treat code as a mathematical solution rather than a software artifact. They prioritize "binary success" (the code runs) over "structural health" (the code is maintainable). For instance, HumanEval does not penalize models for producing "spaghetti code" or failing to include docstrings, provided the output passes the unit tests.

VibeBench was developed to bridge this gap between functional testing and software engineering audits. While existing tools like Evaluating Large Language Models Trained on Code focus on the logic of the algorithm, VibeBench provides an automated pipeline for quantifying technical debt. To the authors' knowledge, VibeBench is among the first frameworks to integrate AST-based heuristic detection for "AI-isms" with Unix-controlled dynamic resource limiting to audit the operational parity of LLM-synthesized software against a formalized human baseline.

Existing static analysis tools such as Pylint and Bandit focus on general Python code quality rather than on patterns specific to LLM-generated outputs, and do not integrate dynamic sandboxed execution or comparison against a human baseline.

Software design

VibeBench is designed as a modular, extensible pipeline written in Python. The framework follows a "Collector-Analyzer-Executor" architecture to ensure that static heuristics and dynamic performance metrics are decoupled and independently verifiable. This separation is deliberate: static analysis operates on source code as text, requiring no execution environment, while dynamic execution requires a controlled subprocess environment with resource limits. By

keeping these two analytical tracks independent, VibeBench allows researchers to run static analysis on any Python file regardless of whether it is safe to execute, and to add new heuristics to either track without affecting the other. The core logic is divided into three primary sub-packages, each with a single well-defined responsibility.

Static Quality Analyzer (`core/analyzer.py`)

The `CodeAnalyzer` class serves as the static analysis engine, utilizing Python's built-in `ast` module to parse generated code into a tree structure without execution.

- **Heuristic Engine:** Implements custom AST visitors to detect "AI-isms" — patterns common in LLM outputs such as ghost comments (empty # symbols) and hardcoded credentials.
- **Complexity Metrics:** Integrates the `radon` and `halstead` libraries to compute Cyclomatic Complexity and Halstead Volume, providing a mathematical basis for maintainability assessment.

Sandboxed Dynamic Executor (`core/executor.py`)

The `CodeExecutor` implements a secure lifecycle management system for safely evaluating unverified AI-generated code.

- **Resource Limiting:** Leverages the Unix `resource` module to enforce hard limits on CPU time (`RLIMIT_CPU`) and maximum memory address space (`RLIMIT_AS`), preventing infinite loops or memory exhaustion from destabilizing the host system.
- **Isolation:** Each test run executes in a clean environment, capturing `stdout`, `stderr`, and exit codes to determine runtime stability.

Reporting and Visualization (`core/reporter.py`)

The reporting layer aggregates JSON-formatted raw data into human-readable outputs.

- **Leaderboard Generation:** Automatically calculates averages across multiple trials and generates Markdown tables for direct inclusion in documentation.
- **Performance Plotting:** Utilizes `matplotlib` to visualize the correlation between structural complexity scores and execution success rates.

Research impact statement

VibeBench enables a category of empirical software engineering research that existing benchmarks cannot support: the systematic audit of LLM-generated code as a software artifact rather than a mathematical solution. By integrating static quality heuristics with sandboxed dynamic execution, VibeBench allows researchers to answer questions that pass/fail unit testing cannot address.

In our initial evaluation across six models and five tasks, VibeBench revealed several findings with direct implications for AI-assisted software development. First, all evaluated models exhibited a significant documentation gap: Claude produced 0% docstring coverage across all tasks despite an 80% functional success rate, demonstrating that functional correctness and code maintainability are independent dimensions that must be measured separately. Second, human-authored baseline solutions achieved lower average cyclomatic complexity (3.60) than every evaluated AI model, confirming a systematic over-engineering tendency in LLM outputs that has practical consequences for long-term code maintenance costs. Third, VibeBench's heuristic detection identified a mutable default argument anti-pattern in DeepSeek's TASK-005 output — a runtime-risk pattern that caused an actual execution failure and that no existing benchmark would surface.

These findings demonstrate that VibeBench fills a genuine measurement gap in the LLM evaluation landscape. Researchers studying model evolution, comparing code generation approaches, or building AI-audit pipelines for production deployment can use VibeBench to quantify dimensions of code quality that are invisible to correctness-only benchmarks. The framework is designed to be extensible, allowing the research community to add new heuristics, models, and task categories as the field evolves.

Mathematics

VibeBench quantifies software quality through two complementary metric families: static complexity measures derived from source structure, and dynamic operational metrics derived from sandboxed execution.

Static Complexity

Halstead Volume (V) measures the information content of a program based on operator and operand counts (Halstead, 1977):

$$V = (N_1 + N_2) \cdot \log_2(n_1 + n_2)$$

where N_1 and N_2 are the total counts of operators and operands respectively, and n_1 and n_2 are the counts of unique operators and operands. Higher volume indicates greater cognitive load required to understand the code.

Cyclomatic Complexity (M) quantifies the number of linearly independent paths through a program's control flow graph (McCabe, 1976):

$$M = E - N + 2P \tag{1}$$

where E is the number of edges, N is the number of nodes in the control flow graph, and P is the number of connected components. VibeBench flags code with $M > 10$ as high-risk for maintainability failure, consistent with McCabe's original threshold.

Dynamic Operational Metrics

Operational Parity (Φ) measures how closely an LLM-generated solution matches the runtime efficiency of a human-authored baseline:

$$\Phi = \frac{T_{\text{base}}}{T_{\text{llm}}} \tag{2}$$

where T_{base} is the execution latency of the human baseline and T_{llm} is the execution latency of the LLM-generated code under identical resource constraints. A value of $\Phi \approx 1$ indicates optimal parity; $\Phi \gg 1$ indicates the LLM solution is significantly slower than the human baseline.

The composite **VibeBench Score** (Σ) aggregates static and dynamic performance into a single normalized metric:

$$\Sigma = w_1 \cdot \hat{V} + w_2 \cdot \hat{M} + w_3 \cdot \Phi$$

where $\hat{V}$ and $\hat{M}$ are min-max normalized Halstead Volume and Cyclomatic Complexity respectively, and w_1, w_2, w_3 are configurable weights summing to 1. Default weights are set empirically at $w_1 = 0.4$, $w_2 = 0.4$, $w_3 = 0.2$.

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Limitations

VibeBench is subject to several limitations that constrain the generalisability of its current findings and should be considered when interpreting results.

First, the benchmark dataset in its current form covers five tasks across three difficulty levels and six models. While sufficient to demonstrate proof-of-concept findings, this scale is modest compared to established benchmarks such as HumanEval (164 tasks) or MBPP (374 tasks). Conclusions about systematic LLM behaviour should therefore be treated as preliminary until validated across a larger task set.

Second, VibeBench currently supports only Python. The "documentation crisis" and over-engineering patterns observed in this study may differ substantially in statically-typed languages such as Java or TypeScript, where type annotations provide an alternative form of documentation and compilers enforce structural constraints that Python does not.

Third, the sandboxed execution environment uses Unix-based resource limits (RLIMIT_CPU and RLIMIT_AS) and is therefore not directly portable to Windows without modification. The static analysis components function cross-platform, but dynamic execution results may vary across operating systems.

Fourth, the Human Baseline used in this study consists of solutions authored by a single developer. A more robust baseline would aggregate solutions from multiple experienced developers to account for individual variation in coding style and complexity.

These limitations represent concrete directions for future work. Expanding the task set, adding multi-language support, and recruiting multiple baseline authors are planned for subsequent releases of VibeBench.

Figures

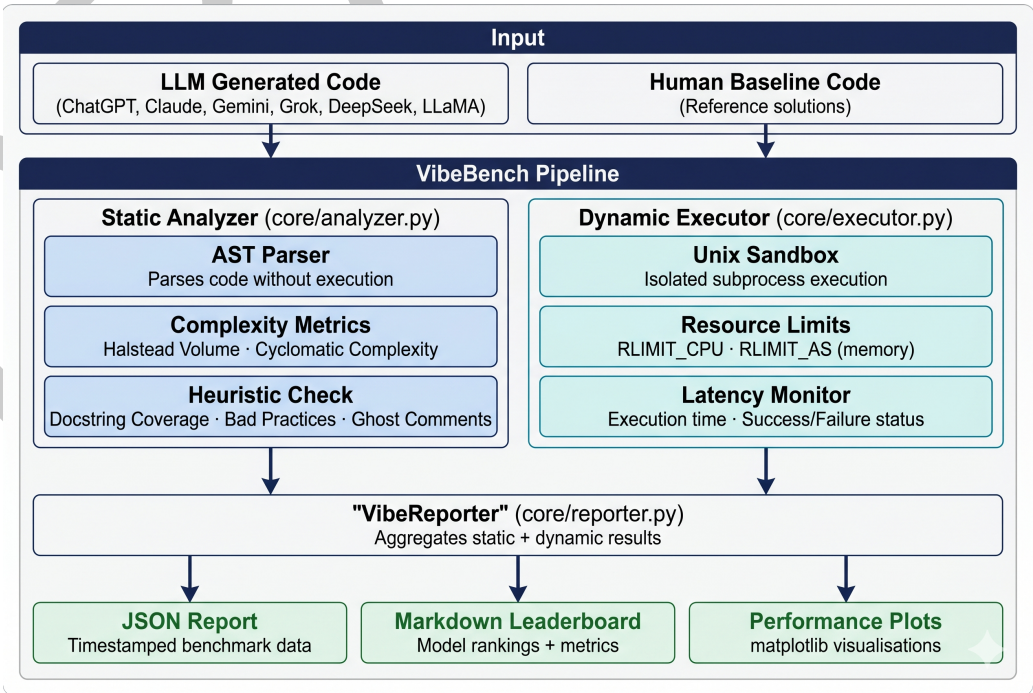

Figure 1: The VibeBench System Architecture, illustrating the modular flow from LLM code ingestion to AST-based static analysis and sandboxed execution.

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As shown in Figure 1, the framework ensures that static heuristics (Halstead complexity, docstring coverage) are captured independently of dynamic performance metrics.

Rank	Model	Avg Complexity	Avg Docstring Coverage	Bad Practices	Success Rate	Avg Execution Time
🥇 1	ChatGPT	4.76	80.0%	0	5/5 (100%)	0.111s
🥈 2	Gemini	3.90	95.0%	0	5/5 (100%)	0.109s
🥉 3	Claude	3.77	0.0%	0	4/5 (80%)	0.131s
4	LLaMA 3.3 70B	5.30	0.0%	0	4/5 (80%)	0.077s
5	DeepSeek	4.65	93.1%	1	2/5 (40%)	0.135s
6	Grok	5.24	82.0%	0	1/5 (20%)	0.161s
—	HUMAN BASELINE	3.60	20.0%	0	5/5 (100%)	0.147s

Figure 2: A sample of the VibeBench Leaderboard output, demonstrating how model performance is ranked across multiple trials.

AI usage disclosure

In accordance with JOSS policies, the author discloses that generative AI tools (specifically Google Gemini and ChatGPT) were utilized during the development of this project.

- **Software Development**: AI was used to assist in writing standard boilerplate code for the reporting engine and for debugging Abstract Syntax Tree (AST) visitor patterns. The core logic of the VibeBench evaluation framework, including the specific static quality heuristics and the dynamic resource-limiting implementation, was designed and verified by the author.
- **Manuscript Preparation**: AI tools were used for language refinement, grammatical correction, and optimizing the LaTeX formatting of the manuscript.

All scientific claims, experimental results, and data interpretations presented in this paper are the original work of the author and have been manually verified for accuracy.

Acknowledgements

The author thanks the faculty and administration of Saint Joseph Higher Secondary School, Dhaka, for fostering an environment that supports independent student research. The author also thanks Mushrat Salehin Nibir for his valuable feedback on the codebase, suggestions for improvement, and for identifying and reporting issues during development. Special thanks are also extended to the Executive Committee and members of the Josephite Scintilla Science Club (JSSC) for their practical feedback on the framework's utility and their continued encouragement of student-led work in software engineering and AI research.

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